Complete Git Commands Guide - From Scratch to Advanced

For Windows Git Bash | Created for your git-journey

1. SETUP & CONFIG - First Time Setup

COMMAND	WHAT IT DOES	EXAMPLE
`git --version`	Check if git is installed and version	`git --version`
`git config --global user.name "Name"`	Set your name for commits (first time)	`git config --global user.name "Shubh"`
`git config --global user.email "email"`	Set your email for commits	`git config --global user.email "shubh@email.com"`
`git config --list`	Show all config settings	`git config --list`
`git config --global core.editor "code --wait"`	Set VS Code as default editor for git	`git config --global core.editor "code --wait"`

2. STARTING A PROJECT - Creating Repo

COMMAND	WHAT IT DOES	EXAMPLE
`git init`	Initialize a new git repo in current folder. Creates .git folder	`git init`
`git clone`	Clone/download existing repo from GitHub to your PC	`git clone https://github.com/user/repo.git`
`git clone myFolder`	Clone into a specific folder name	`git clone https://github.com/user/repo.git myFolder`

3. BASIC WORKFLOW - Daily Commands

COMMAND	WHAT IT DOES	EXAMPLE
`git status`	Show status of files: tracked, untracked, modified. MOST USED	`git status`
`git add`	Add single file to staging area	`git add three.txt`
`git add .`	Add ALL files in current folder to staging	`git add .`
`git add -A`	Add all files even outside current folder	`git add -A`

git reset	Remove file from staging (unstage)	`git reset three.txt`
git commit -m "message"	Save staged changes to local repo with message	`git commit -m "added three.txt"`
git commit -a -m "message"	Add + commit modified files in one step (not new files)	`git commit -a -m "update"`
git log	Show history of commits. Press q to exit	`git log`
git log --oneline	Show log in one line per commit (short)	`git log --oneline`
git log -10	Show last 10 commits	`git log --oneline -10`
git diff	Show changes not yet staged	`git diff`
git diff --staged	Show changes that are staged but not committed	`git diff --staged`

4. FILE MANAGEMENT

COMMAND	WHAT IT DOES	EXAMPLE
touch file.txt	Create new empty file (Linux/Git Bash)	`touch three.txt`
git rm	Delete file from git and disk	`git rm three.txt`
git rm --cached	Remove file from git but keep on disk	`git rm --cached three.txt`
git mv old new	Rename file in git	`git mv old.txt new.txt`
cat	Show content of file	`cat three.txt`

5. BRANCHING - Most Important for Jobs

COMMAND	WHAT IT DOES	EXAMPLE
git branch	List all local branches	`git branch`
git branch -a	List all local + remote branches	`git branch -a`
git branch	Create new branch	`git branch feature-login`
git checkout	Switch to another branch (old way)	`git checkout main`
git switch	Switch to branch (new way, better)	`git switch main`
git checkout -b	Create and switch to new branch in one	`git checkout -b feature-login`
git switch -c	Create and switch (new way)	`git switch -c feature-login`
git branch -d	Delete branch (safe)	`git branch -d feature-login`
git branch -D	Force delete branch	`git branch -D feature-login`

git branch -m newName	Rename current branch	`git branch -m main`

6. MERGING & CONFLICTS

COMMAND	WHAT IT DOES	EXAMPLE
git merge	Merge branch into current branch	`git merge feature-login`
git merge --abort	Cancel merge if there are conflicts	`git merge --abort`
git checkout --conflict=diff	Show conflict details	`git checkout --conflict=diff file.txt`

7. CONNECTING TO GITHUB - Remote

COMMAND	WHAT IT DOES	EXAMPLE
git remote	Show remote names	`git remote`
git remote -v	Show remote URLs (detailed)	`git remote -v`
git remote add origin	Connect local repo to GitHub repo	`git remote add origin https://github.com/user/repo.git`
git remote remove origin	Disconnect from GitHub	`git remote remove origin`
git push -u origin main	First push + set upstream	`git push -u origin main`
git push	Push commits to GitHub (after first)	`git push`
git pull	Pull/download latest changes from GitHub	`git pull`
git pull origin main	Pull from specific branch	`git pull origin main`
git fetch	Download changes but don't merge yet	`git fetch`

8. UNDO & FIX MISTAKES - Very Important

COMMAND	WHAT IT DOES	EXAMPLE
git restore	Discard changes in file (unstaged)	`git restore three.txt`
git restore --staged	Unstage file	`git restore --staged three.txt`
git checkout --	Discard changes (old way)	`git checkout -- three.txt`
git reset --soft HEAD~1	Undo last commit, keep changes staged	`git reset --soft HEAD~1`
git reset --mixed HEAD~1	Undo last commit + unstage (default)	`git reset --mixed HEAD~1`

git reset --hard HEAD~1	Undo last commit + delete changes (DANGER)	`git reset --hard HEAD~1`
git revert	Create new commit that undoes old commit (safe for team)	`git revert 923bc72`
q	Exit log view / less view	`press q`

9. STASH - Save Work Temporarily

COMMAND	WHAT IT DOES	EXAMPLE
git stash	Save current changes temporarily	`git stash`
git stash list	List all stashes	`git stash list`
git stash apply	Bring back last stash	`git stash apply`
git stash pop	Bring back and delete stash	`git stash pop`
git stash clear	Delete all stashes	`git stash clear`

10. ADVANCED - For Interviews & Production

COMMAND	WHAT IT DOES	EXAMPLE
git tag v1.0	Create version tag	`git tag v1.0`
git tag	List tags	`git tag`
git push origin --tags	Push tags to GitHub	`git push origin --tags`
git cherry-pick	Copy a specific commit to current branch	`git cherry-pick 923bc72`
git rebase main	Replay your branch on top of main (clean history)	`git rebase main`
git rebase --abort	Cancel rebase	`git rebase --abort`
git blame	Show who changed each line	`git blame data.py`
git show	Show details of a commit	`git show 923bc72`
git shortlog	Show commits grouped by author	`git shortlog`
explorer .	Open current folder in Windows (not git, but needed)	`explorer .`
start .	Also opens folder in Windows	`start .`
code .	Open folder in VS Code	`code .`

Quick Cheat Sheet - Daily Workflow

Every Day Flow:

1. git pull (get latest)
2. Do your work, edit files
3. git status (check)
4. git add . (stage)
5. git commit -m "clear message" (save)
6. git push (send to cloud)

Your Problem Solutions:

- Can't paste HTTPS? -> Right Click -> Paste or Shift+Insert
- Stuck in git log? -> Press q
- open . not working on Windows? -> Use explorer . or start .
- bash: q: command not found? -> You typed $ q, just press q alone without $
- ((923bc72...)) showing? -> You are in detached HEAD, do git checkout main